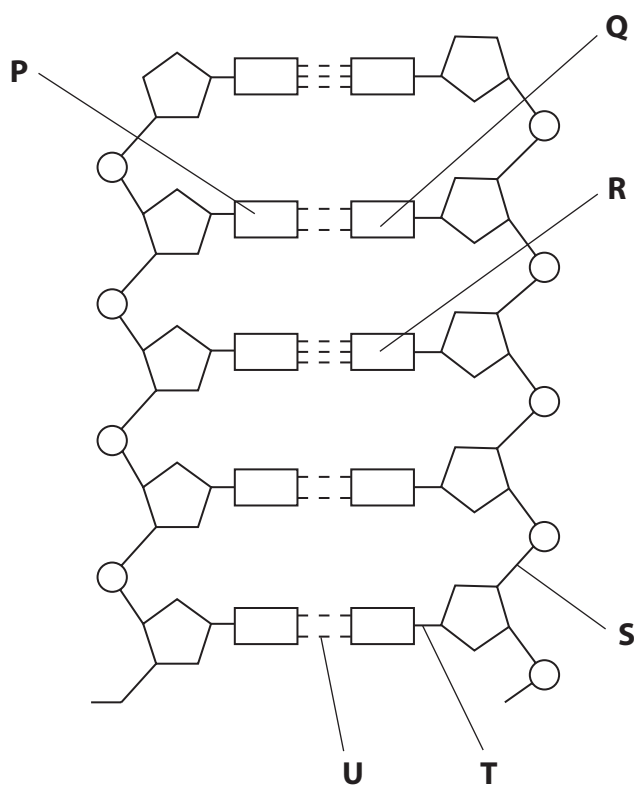


4 The polynucleotide DNA is composed of mononucleotides linked together.

Two polynucleotides form a DNA molecule.

(a) The diagram shows part of a DNA molecule.



(i) Draw a circle around **one** mononucleotide that includes the base labelled **R**.

(1)

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(ii) Which row of the table identifies the bonds labelled **S**, **T** and **U**?

(1)

|                                   | <b>S</b>       | <b>T</b>       | <b>U</b>       |
|-----------------------------------|----------------|----------------|----------------|
| <input type="checkbox"/> <b>A</b> | hydrogen       | phosphodiester | covalent       |
| <input type="checkbox"/> <b>B</b> | hydrogen       | covalent       | phosphodiester |
| <input type="checkbox"/> <b>C</b> | phosphodiester | hydrogen       | covalent       |
| <input type="checkbox"/> <b>D</b> | phosphodiester | covalent       | hydrogen       |

(iii) The base labelled **P** is adenine.

Which is the base labelled **Q**?

(1)

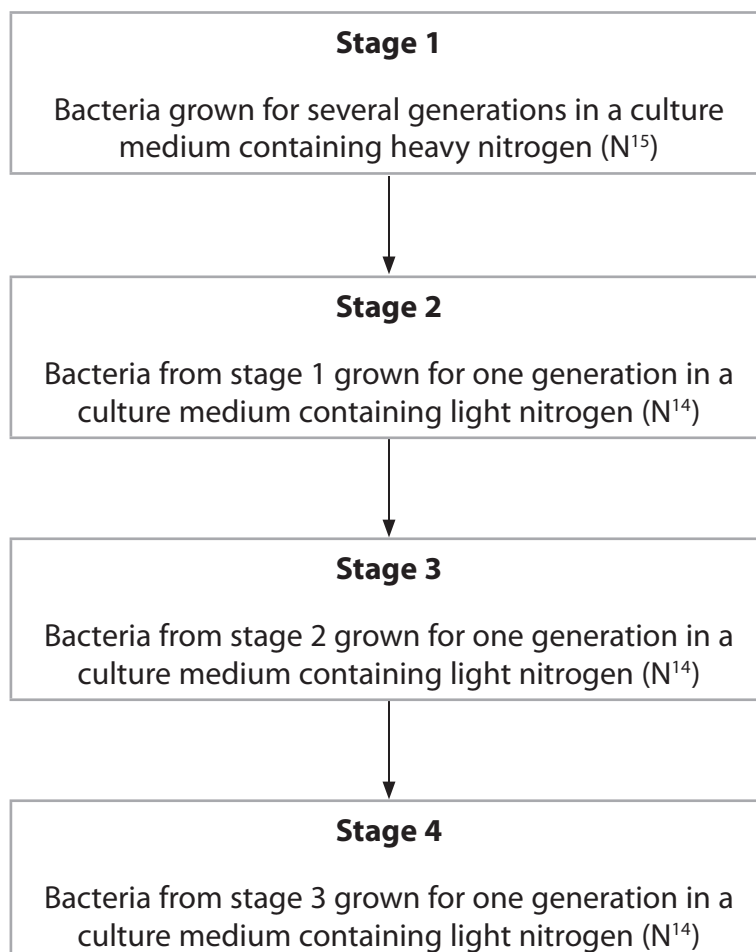
- ☐ **A** cytosine
- ☐ **B** guanine
- ☐ **C** thymine
- ☐ **D** uracil



- (b) Meselson and Stahl carried out experiments that provided evidence for the semi-conservative replication of DNA.

Heavy nitrogen ( $N^{15}$ ) and light nitrogen ( $N^{14}$ ) were used in these experiments.

The flow chart summarises part of one experiment performed by Meselson and Stahl.



After each stage, a sample of DNA was taken from the bacteria and the DNA molecules separated using a density gradient in a tube.

The heavier DNA molecules form bands lower down the gradient than the lighter DNA molecules.

The height of each band is proportional to the percentage of DNA molecules in the sample.



(i) Complete the diagram to show the results of this experiment.

Use the dotted lines to help you to position the bands on the diagram.

The first one has been done for you.

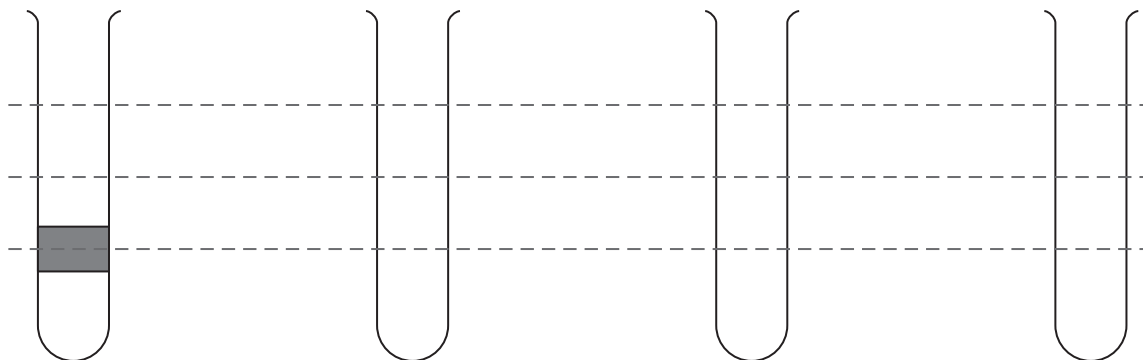
(5)

DNA sample taken  
after stage 1

DNA sample taken  
after stage 2

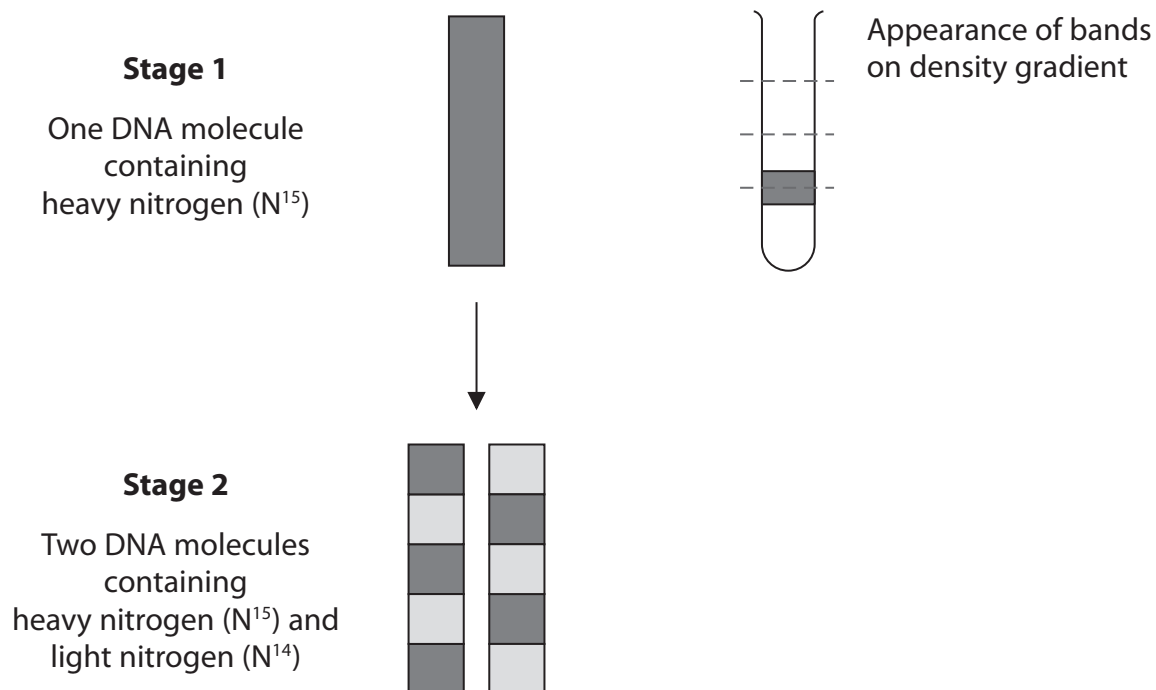
DNA sample taken  
after stage 3

DNA sample taken  
after stage 4



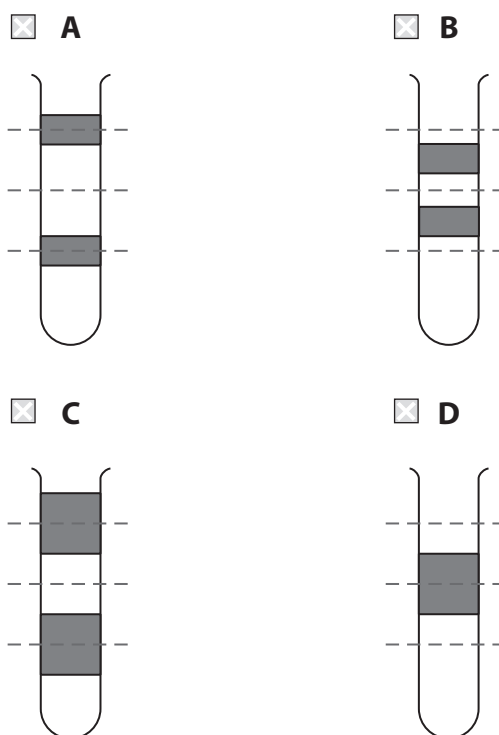
- (ii) The experiments of Meselson and Stahl disproved the dispersive theory of DNA replication.

The diagram shows the expected results if the dispersive theory was correct.



Which diagram would show the bands of DNA molecules on the density gradient at stage 2, if the dispersive theory was correct?

(1)



(Total for Question 4 = 9 marks)

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